



Lesson Aim

Investigate unique processes used to make red and rosé wines.

INTRODUCTION

The biggest difference between red and white wine production is that red wines are fermented with skins still present.

It is worth understanding that the juice of even many red grapes is clear. It is the skin that holds the colour, various flavours and the tannins that give a red wine its taste. Stalks and pips can also add to that uniquely red wine flavour, however excessive stalks and pips can negatively affect the flavour.

Red wines most commonly only use red or black grapes. The amount of colour in a red wine depends on both the grape variety used, and the length of time that the must is exposed to the skins. The action of yeast and alcohol during fermentation will extract colour from skins.

- Stems may or may not be removed when grapes are crushed.
- Grapes go into the fermenting tank with the skins (and sometimes stems).
- Wine which runs off freely is put into barrels and allowed to mature for varying periods.
- Skins and remaining moisture in the vat can be pressed to retrieve further dark and bitter wine; this can be blended with the wine which was run off earlier to give differing tastes.

HARVESTING AND CRUSHING

Fine quality red wines are more commonly made from hand harvested grapes. Mechanical harvested grapes can be used to make perfectly adequate red wines, but there is greater opportunity to control aspects of the process with hand harvesting.

Oxidation is less of an issue with red and black grapes than with white grapes; nevertheless, methods to control oxidation in white wine making are also commonly used for reds as well.

Crushed red grapes have been fermented frequently in some parts of the world, without adding sulphur dioxide, however adding sulphur dioxide at the point of crushing is commonly recommended.

FERMENTATION

Red wines are more commonly fermented in open tanks than whites. This is because oxidation is not as much of a problem, combined with the fact that skins tend to float on the surface creating a partial barrier to the must. In addition, any slightly undesirable flavours that develop in a red wine are more likely to be undetectable because red have such a strong taste.

Contamination is nevertheless possible, particularly at the start and finish of fermentation; hence lids are commonly used at least early and late in the fermentation process.

The layer of skins floating on the surface of fermenting must may protect it from oxidation, but it can also have a negative effect. As carbon dioxide is produced and rises to the surface it can be trapped between the must and the skins. This situation can then reduce the transfer of colour into the red must. Skins might also trap air or undesirable microorganisms. Agitating the layer of skins regularly (e.g., every 5 to 7 hours) can overcome these problems. On a small scale, vats of wine might be stirred by hand. On a larger scale machinery might be used to agitate it.

Red wines are commonly fermented at temperatures higher than white wines, ideally 18-25°C. Some more strongly flavoured red wines are fermented at higher temperatures. Pinot Noir for instance may be fermented at temperatures as high as 35°C. High temperature fermentation is much faster though and needs to be monitored closely; potentially taking as little as only two weeks

High temperatures help extraction of colour do increase oxidation and release of tannins and polyphenols. Flavour tends to be different at higher fermentation temperature also. The ratio of skin to juice in the must is a further factor that affects colour. All these factors need to be managed and balanced by the wine maker.

Once sufficient colour has leached from the skins, the must should be moved to a press. Pressing then allows for skins and other particles to be removed. The clarified must if still fermenting, is then moved to a sealed fermenting tank/vat, and fermentation can progress. In cooler climates the fermentation may be complete or near to complete at pressing. If wine is fermented fully with skins still in the must, it can be critical to press and remove skins immediately once fermentation ends. A variation on this approach, called maceration, is traditionally used for making some wines including some clarets. Once wine has been pressed and skins removed, ongoing lower temperature fermentation can occur at 10-18°C.

EXTRACTING COLOUR

Colour extraction is used when making red and rosé wines, but not with whites.

Mashing

Mashing involves maceration before fermenting rather than during fermentation. This process can greatly increase the speed of colour extraction, allowing an earlier pressing, and either the same or a darker final colouring.

Storing on Skins

This involves crushed grapes being stored for a time in a sealed container under a layer of carbon dioxide. This can be 10 to 14 weeks at 18- 24°C. After colour has been achieved, the pressing is carried out, and skins removed before fermentation.

Maceration

The process of winemaking is where the grape phenolic materials like tannins, colouring agents (anthocyanins) and flavour compounds are leached from the grape skins, seeds and stems into the must. This involves plunging well before all the sugar is used up in the fermentation process. It is a process that can help produce a "softer" red wine. For an authentic Beaujolais style wine, grapes have stems removed, are crushed, and are then fermented to produce a lighter coloured, fresher more aromatic flavour.

Carbonic Maceration

It consists of fermenting whole berries in a CO₂ saturated atmosphere without being crushed. In this process, whole clusters fill the tank or any other container containing CO₂. The principle is to cover all the fruit with CO₂ to establish an anaerobic environment. Some winemakers put fermenting must (about 5 to 10 per cent), which produces CO₂ in the bottom of the tank.

The tank is loaded with the fruit and sealed. Partial fermentation starts inside the cells under anaerobic conditions. The cell's own enzymes cause this fermentation (without yeast). A small amount of about 1.5 to 2.5% of alcohol is created by fermentation, resulting in several changes in the composition of the must.

The fermentation takes place for about eight to ten days and the temperature is held close to 35°C. The clusters are removed and pressed after this whole berry fermentation. The partially fermented juice is inoculated with wine yeast and the fermentation is complete.

Due to lower phenolics and decreased acidity, wines so produced are softer and have a characteristic fermentation aroma. These wines are clarified, stabilized, finished, and offered for consumption within a few months of the vintage.



Short Macerations

Pink wines are created by macerating red wine juice on the skins for a very short time period. It is a perfect way to make red wine grapes that can be consumed even more easily than a full-fledged red wine. There is only short time for wine to pick up a little colour and some spice, but very little tannins are removed.

Cold Soak

The pre-fermentation maceration also known as cold soak as it involves chilling of the must below 12°C to inhibit yeast fermentation. At extraction at this temperature will move slowly, giving you much control over how much you extract.

The grape solids and stems should be extracted once the optimum extraction has been achieved and the must be brought back up to a higher temperature so that fermentation can begin. You can allow wild yeast from the grape skins to get things going at that stage or inoculate your own yeast.

Thermovinification

It is a pre-fermentation heating technique used to promote rapid extraction of colour from skins. Red wines produced using thermovinification tend to have a softer and lighter body, less astringency and often better colour, compared to traditionally made red wines. Wines made this way also tend to need less aging. Some types of wine grape (including Pinot types) are more suited to this method than others.

Method 1

Involves heating the must to as high as 60, after stalks are removed and the grapes have been crushed. The must be kept at that temperature for up to 3 hours then cooled between 10- 12°C. Once cooled, 50ppm of sulphur dioxide is added (Never add sulphur dioxide before or during heating). Once the must has been coloured this way, it can then be pressed, the skins removed, and it can then be fermented in much the same way as is done with white wine.

Method 2

This involves using heating that increases the pressed must by around 80°C. This method only needs one to two minutes to extract the colour from the skin.

HOW TO GET A WOOD FLAVOUR

Traditionally red wines are aged in oak casks (barrels) to add a subtle wood flavour, but oak casks are expensive. Oak comes from primarily from France and United States (French and American oak).

Another technique that has been occasionally used is adding wood chips or shavings to the wine. One authority has reported success from adding oak wood chips to must while fermenting with the skins, then removing chips at the same time as removing skins. Amounts of oak wood chips may vary and should be carefully considered.

